treatment in which the gravitational contribution to entropy is cleanly identified, separated off, and accounted for. Nevertheless, this doesn't compromise the qualitative conclusions we reach. In case you find this troublesome, note that the whole discussion can be rephrased largely without reference to gravitational entropy. As we emphasized in Chapter 6, when ordinary attractive gravity is relevant, matter falls together into clumps. In so doing, the matter converts gravitational potential energy into kinetic energy that, subsequently, is partially converted into radiation that emanates from the clump itself. This is an entropy-increasing sequence of events (larger average particle velocities increase the relevant phase space volume; the production of radiation through interactions increases the total number of particles—both of which increase overall entropy). In this way, what we refer to in the text as *gravitational entropy* can be rephrased as *matter entropy generated by the gravitational force.* When we say gravitational entropy is low, we mean that the gravitational force has the potential to generate significant quantities of entropy through matter clumping. In realizing such entropy potential, the clumps of matter create a non-uniform, non-homogeneous gravitational field—warps and ripples in spacetime—which, in the text, I've described as having higher entropy. But as this discussion makes clear, it really can be thought of as the clumpy matter (and radiation produced in the process) as having higher entropy (than when uniformly dispersed). This is good since the expert reader will note that if we view a classical gravitational background (a classical spacetime) as a coherent state of gravitons, it is an essentially unique state and hence has low entropy. Only by suitably coarse graining would an entropy assignment be possible. As this note emphasizes, though, this isn't particularly necessary. On the other hand, should the matter clump sufficiently to create black holes, then an unassailable entropy assignment becomes available: the area of the black hole's event horizon (as explained further in Chapter 16) is a measure of the black hole's entropy. And this entropy can unambiguously be called gravitational entropy.

5. Just as it is possible both for an egg to break and for broken eggshell pieces to reassemble into a pristine egg, it is possible for quantum-induced fluctuations to grow into larger inhomogeneities (as we've described) or for sufficiently correlated inhomogeneities to work in tandem to suppress such growth. Thus, the inflationary contribution to